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10 December 1981
SUPERSEDING
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22 December 1967

MILITARY SPECIFICATION

# TRAILER, BOLSTER, POLE TYPE: 14-TON, 4-WHEEL, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 Scope. This specification covers a four-wheel, tandem axle, 14-ton capacity, pole type bolster trailer with a second bolster for mounting on the towing vehicle. Vehicles produced under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS
MILITARY

* MIL-STD-1223 - Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2330

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MS51336 - Lunette Coupler, Drawbar, Ring.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

* DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* SAE, INC.
SAE Standards and Recommended Practices.

| | | |
|---|---|---|
| * | J318 | - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers. |
| | J560 | - Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable. |
| * | J588 | - Turn Signal Lamps. |
| | J697 | - Safety Chain of Full Trailers or Converter Dollies. |
| | J849 | - Connection and Accessory Locations for Towing Double Trailers and Multi-Axle Trailers. |
| * | J1067 | - Seven Conductor Jacketed Cable for Truck and Trailer Connectors. |
| * | J1292 | - Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring. |

* (Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.



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(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.4, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, which meet or exceed the requirements of this specification. All vehicle items shall be as represented in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting, marking, and data plates. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223.

3.1.1.3 Drain plugs. All drain plugs installed throughout the vehicle for the purpose of draining lubricants shall be of the permanent magnet type.

3.1.1.4 Lunette. The lunette shall be in accordance with MS51336 (see 3.4.6).



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3.2 Vehicle weights and ratings.

3.2.1 Curb weight. The curb weight of the fully equipped vehicle shall include chassis, bolsters, and all attachments and accessories.

* 3.2.2 Payload. The payload capacity shall be not less than 28,000 pounds, concentrated at the transverse centerline of the rear bogie.

* 3.2.3 Gross weight. The gross weight shall include the curb weight of the vehicle and the specified payload (see 3.2.2). The imposed loading on the axles, measured at the ground, shall be computed using the curb weight and the specified payload concentrated at the transverse centerline of the rear bogie.

* 3.2.4 Ratings. Vehicle ratings shall be the manufacturer's published ratings. The component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verifications of ratings must be submitted to the engineering office of the procuring activity.

* 3.2.5 Dimensions. The trailer dimensions, with the load area level and with the telescopic towbar fully retracted, shall conform to the dimensions specified in table I.

TABLE I. Dimensions.

| Characteristic | Dimension |
|---|---|
| Overall length, maximum, from centerline of lunette to rear of trailer (inches) | 330 |
| Wheelbase, maximum, from centerline of lunette to centerline of bogie (inches) | 260 |
| Overall width (plus or minus 1 inch) | 95 |

* 3.3 Performance. The trailer performance as specified in 3.3.1 through 3.3.4 shall be met by the trailer when fully equipped, completely serviced, and towed by a towing vehicle over improved and unimproved, hard, even terrain. Performance requirements shall be met with the towbar fully extended, fully retracted, or in any intermediate position, with the trailer fully loaded and with the trailer empty. The trailer shall show no evidence of failure or malfunction of components, attachments, or accessories.

* 3.3.1 Travel speed. The trailer shall be capable of being towed over dry improved gravel roads at speeds as great as 15 miles per hour (mph), and shall be capable of being towed over improved roads at speeds as great as 55 mph.



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* 3.3.2 Turning ability. The trailer, when coupled to a 96 inch wide towing vehicle, shall be capable of forming a 70 degree angle with the towing vehicle, without interference and without damage to the trailer or the towing vehicle.

* 3.3.3 Trailing ability. The trailer shall not exceed the allowance for tracking deviation specified in Federal Motor Carrier Safety Regulation 393.70(a).

* 3.3.4 Service brake performance. The service brakes of the trailer and towing vehicle shall control, decelerate and stop the vehicle combination in accordance with Federal Motor Carrier Safety Regulation 393.52.

3.4 Trailer components.

3.4.1 Frame. The chassis frame members shall be adequately braced with sufficient number of cross members to insure stability. When the trailer is loaded with specified payload (see 3.2.2), the maximum fiber stress in any frame member shall not be greater than 50 percent of the yield strength of the materials used. All parts of the frame shall be completely electric-arc welded. Where frame members are spliced, joints shall be reinforced to fulfill the dynamic and static loading requirements specified herein. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous.

3.4.2 Suspension. The trailer shall be equipped with manufacturer's standard spring suspension system of the walking beam type. Each component shall have a rated capacity at least equal to the respective imposed loads, measured at the ground, with trailer loaded with rated payload as specified in 3.2.2.

3.4.2.1 Springs. The clearances between spring and spring stops shall be sufficient to prevent bottoming. A sufficient number of alinement clips shall be provided to preclude lateral shifting.

3.4.3 Tandem axles. Tandem axles shall be furnished. Through-shaft type axles shall be provided. The axle system shall provide sufficient articulation to maintain equal loading on each wheel. The axle rating shall be at least equal to the load imposed on each axle, measured at the ground with trailer loaded with rated payload as specified in 3.2.3. Articulation shall be demonstrated under all of the following conditions:



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(a) With two diagonally opposite wheels elevated 6 inches
(b) With forward pair of wheels elevated 8 inches above the rear wheels
(c) With rear wheels elevated 8 inches above the forward wheels.

* 3.4.4 Wheels, rims, tires and tubes. The vehicle shall be equipped with four dual wheels. The rim and tire ratings shall conform to The Tire and Rim Association recommendations for the type and size of tire furnished. The tire and rim sizes shall be the same for all wheels on the vehicle. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.4.1 Tires. Tires shall be steel belted radial ply or, when specified (see 6.2), bias ply. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded with rated payload.

3.4.4.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type, and shall be of proper size for tire furnished. Tire flaps shall be provided for tube type tires in accordance with The Tire and Rim Association recommendations.

* 3.4.5 Brakes. Full air, internal expanding type brakes shall be furnished. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40, 393.42, 393.43, 393.45 through 393.48, and 393.52. The brake system shall include piping, detachable coiled air hose lines with gladhands on each end, fixed gladhands, spring loaded dust covers or dummy couplings equipped with security chains for the fixed gladhands, provisions for stowing dummy couplings, relay emergency valve, air reservoirs with a remote control drain located above the bottom of the frame rails, slack adjusters, brake chambers, and all other parts and connections required for a complete air brake system. All gladhands shall conform to SAE J318. All components of the brake system shall be installed in a manner to provide road clearance for travel over uneven or rough terrain and protection from objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of the wheel rims. The air hose lines shall be of sufficient length to meet the turning requirements of 3.3.2 when the telescopic towbar is fully extended. Hangers shall be provided to retain the air hose lines on the towbar, with the towbar both fully extended and fully retracted.

* 3.4.5.1 Parking brakes. The trailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall be automatically applied upon disconnection of the



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emergency air line and under emergency conditions. With no additional application the parking brakes shall hold the trailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.4.5.1.1 Chock blocks and cradles. Four pairs of wood, steel or aluminum chock blocks with corrosion resistant chain and two cradles shall be furnished. The hook of each captive chain shall be secured to the cradle. The captive chains shall be of a length to be wrapped around the cradle to hold the chock blocks in place during travel and allow the wheels to be chocked in front and rear. The chock blocks shall be of sufficient size to prevent vehicle movement during loading and unloading operations, and while parked up or down a 30 percent incline.

* 3.4.6 Telescopic towbar. A rigid type towbar shall be furnished. The towbar shall be provided with a single-tongue telescopic towbar with lunette. The towbar shall be of all steel construction. The telescopic towbar shall be not less than 20 feet in length when retracted and shall be provided with not less than 8 equally spaced stops providing for extension to not less than 40 feet in length. When pins or other removable devices are provided for stops, they shall be secured with corrosion resistant fasteners or chains. The lunette shall be in accordance with MS51336. The towbar shall be in accordance with Federal Motor Carrier Safety Regulation 393.71(h). The towbar shall be essentially in a longitudinal horizontal plane when coupled to a towing pintle that is located in accordance with SAE J849. The towbar shall be of sufficient length and angularity to meet the requirements of 3.3.2 and 3.3.3. The lunette shall be mounted to provide a vertical range of adjustment, measured at the centerline of the lunette, of not less than 4-inches. The towbar and all its attachments shall withstand a 60,000 pound towbar pull without residual deformation.

* 3.4.6.1 Safety chains. Safety chains shall be furnished for the telescopic towbar. The safety chains shall be in accordance with SAE J697 and in accordance with Federal Motor Carrier Safety Regulations 393.70(f) and 393.71(h).

* 3.4.7 Rear end protection. Rear end protection shall be furnished and shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.8 Lashing "D" rings. Two lashing "D" rings shall be provided on each side of the trailer. The lashing "D" rings shall be within the maximum overall vehicle width of 96 inches. Each ring and attachment



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shall withstand a vertical load of 8,000 pounds. Lashing rings shall also be positioned to enable the trailer to be lifted in its normal horizontal position without payload.

* 3.4.9 Electrical system. An electrical lighting system of 12-volt potential shall be provided. The wiring shall be in accordance with SAE J1292 and all wiring exposed to possible physical damage shall be protected by the use of conduit. All wiring shall be color coded in accordance with SAE J560.

* 3.4.9.1 Lighting. The lighting system shall conform to Federal Motor Carrier Safety Regulations 393.16 (including identification lamps on the rear of the pole trailer), 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33. All lights and reflectors shall be mounted in recessed settings or in guarded locations to provide protection from operational hazards. The turn signal lamps shall be in accordance with SAE J588.

* 3.4.9.2 Lighting receptacle and cable. The vehicle shall be equipped with a lighting receptacle mounted in a readily accessible location. The receptacle shall be in accordance with SAE J560 with a spring loaded cover assembly. A coiled electrical cable conforming to SAE J1067 shall be furnished. Both ends of the cable shall be equipped with a seven-circuit round plug conforming to SAE J560. Plugs shall be equipped with a grip for withdrawing from the vehicle connector socket. The cable shall be of sufficient length to meet the requirements of 3.3.2 when the telescopic towbar is fully extended. Hangers shall be provided to retain the cable, with the towbar both extended and fully retracted.

3.5 Body.

* 3.5.1 Bolsters. Two swivel type bolsters shall be furnished. One shall be mounted over the transverse centerline of the rear bogie. The other bolster shall be provided with mounting devices to mount the bolster on the bed of the towing vehicle. A means of securing the bolsters in unloading position shall be provided. The bolsters shall extend the full width of the trailer. The bolster swivel surfaces shall be provided with a means for lubrication and shall be accessible for service.

3.5.2 Stanchions. Two adjustable stanchions shall be furnished on each bolster. The stanchions shall be adjustable for load width and shall be demountable. The stanchions shall be provided with corrosion resistant security chains and cam-type locking devices.



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3.5.3 Accessories. Two of each of the following accessories, for securing the payload on the bolsters, shall be furnished:

(a) A pole binder, equipped with forged steel hooks and eccentric lever to tighten and loosen the pole binder
(b) A chain of not less than 16,000 pound breaking strength for the pole binder, with a slip ring on one end and a grab hook on the opposite end of the chain.

3.6 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: inflation of all tires; adjustment of brakes; and complete lubrication of chassis and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.7 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts and castings shall be commensurate with sound manufacturing standards for similar equipment. The vehicle shall be free from all burrs, sharp edges, rust, scale, and cracks. Improper manufacturing or assembling practices, defective components or parts and assemblies, which have been modified to overcome deficiencies shall not be furnished.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.



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* 4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on the lunette and rear axle. The imposed loading on the lunette and axle shall be computed using the curb weight and the specified payload (see 3.2.2), distributed as specified in 3.2.3. The calculated imposed loads on the axles will be utilized to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contractual requirements.

* 4.3.2 Road test. The first production vehicle shall be examined and road tested (with payload) for 20 miles by the contractor, under the direction and surveillance of Government representatives, to assure that the vehicle will operate in accordance with contract requirements. The vehicle, loaded with the equipment specified herein, shall be towed for a minimum distance of 20 miles. The distance traveled shall consist of: 10 miles over dry, level, smooth, prepared hard-surfaced roads at sustained speeds of 55 mph; and 10 miles over dry improved gravel roads at sustained speeds up to 15 mph. Upon completion of the road test, all equipment shall be thoroughly examined and inspected. The equipment shall show no evidence of damage.

4.3.3 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.4 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the



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point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

* 6.1 Intended use. The vehicles covered by this specification are intended for nontactical use by the Government for transporting poles for overhead cable operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Concealed marking, if required (see 3.1.1.1).
* (d) Rustproofing, if required (see 3.1.1.2).
* (e) Disc type wheels, if required (see 3.4.4).
* (f) Bias ply tires, if required (see 3.4.4.1).

6.3 Identification of changes. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-62083A(AT), TRAILER, BOLSTER, POLE TYPE: 14-TON, 4-WHEEL, COMMERCIAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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